आरंभ BATCH

L-2

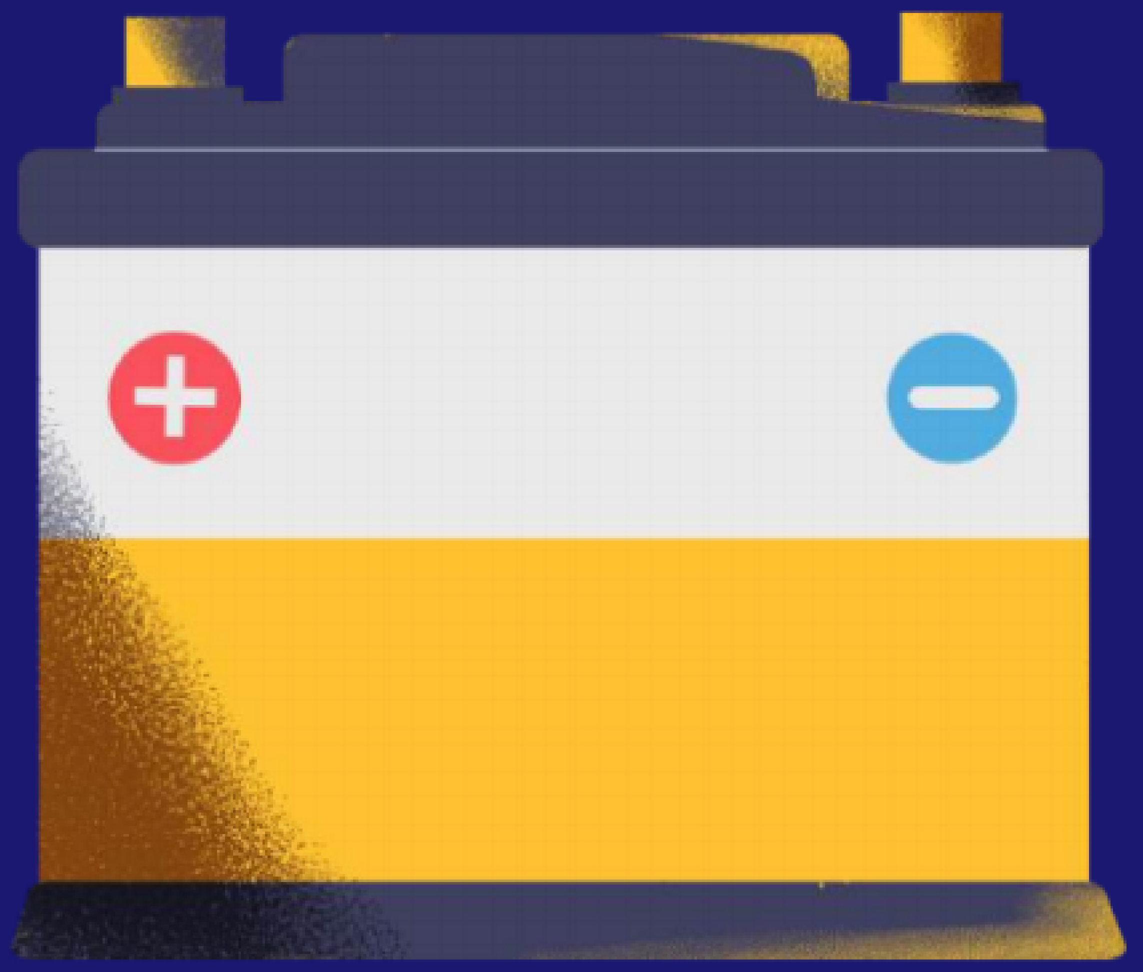
CLASS X – SCIENCE

ELECTRICITY

PRASHANT KIRAD

Role of Cell or Battery

- A cell or battery produces potential difference across its terminals.
- This potential difference pushes the charges and makes them move through the conductor.
- Due to this movement of charges, electric current is produced.
- The cell uses its stored chemical energy to maintain the flow of current in the circuit.



POTENTIAL DIFFERENCE

Electric potential difference between two points is defined as the work done to move a unit charge from one point to another.

+

$$V = W/Q$$

$V = \frac{W}{Q}$ $\frac{J}{C}$

$V \rightarrow J/C$

The SI unit of potential difference is volt, written as V.

Potential difference between two points is said to be 1 volt if 1 joule of work is done to move 1 coulomb of charge from one point to another.

$$V = 1\ JC^{-1}$$

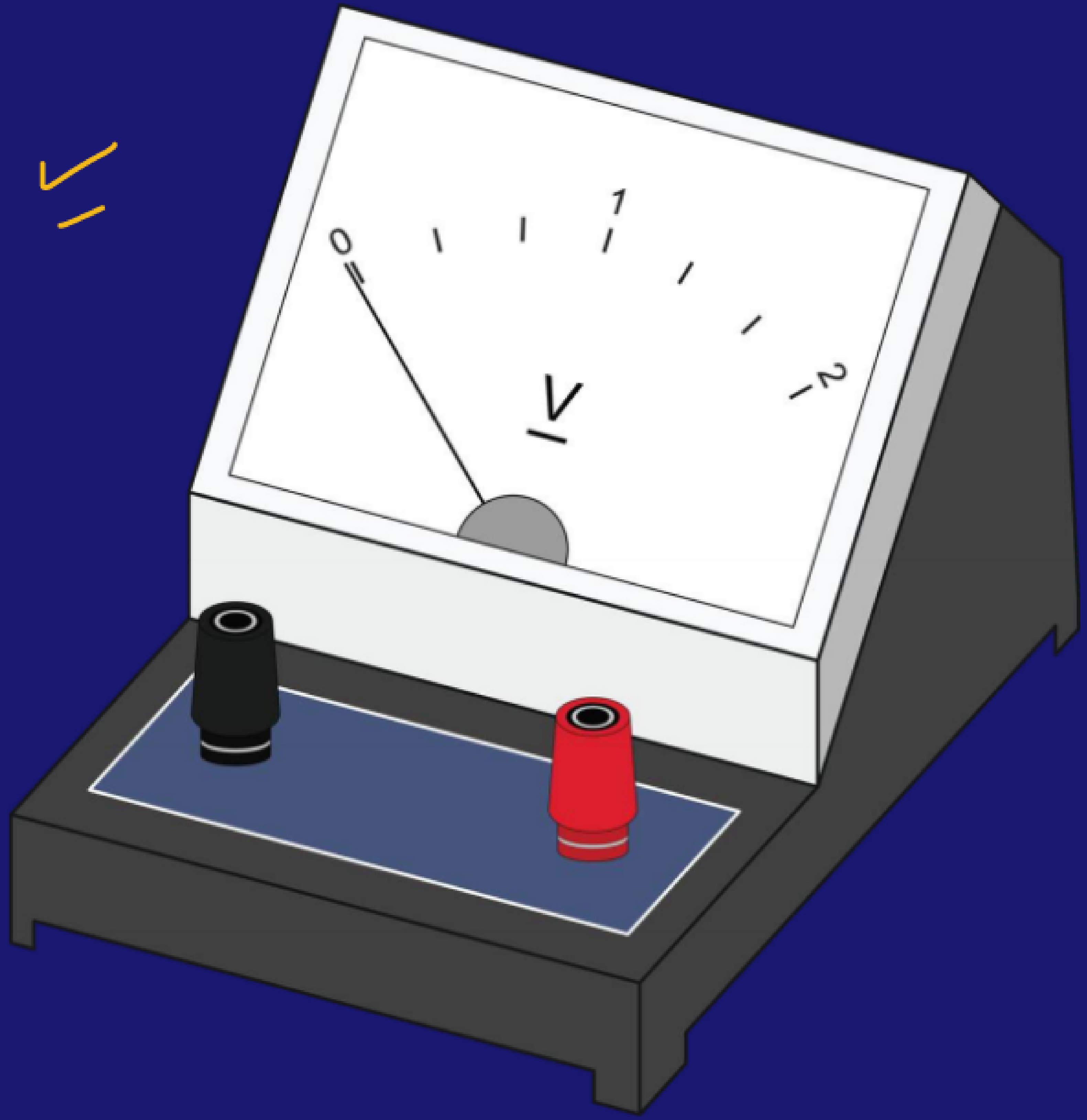
$$V = \frac{W}{Q}$$

$$1V = \frac{1J}{1C}$$

$$1V = \frac{1J}{1C}$$

Voltmeter:

- Potential difference is measured by an instrument called a voltmeter.
- A voltmeter is always connected in **parallel** across the points between which potential difference is to be measured.



Q. The work done in moving a charge from one point to another is 20 J. If the potential difference between the points is 10 V, what is the charge?

$W = 20\,J$

$V = 10\,V$

$$V = \frac{W}{Q}$$

$$Q = \frac{W}{V} = \frac{20}{10} = 2\,C$$

[NCERT]

Q. How much work is done in moving a charge of 2 C across two points having a potential difference 12 V?

$V = 12V$

$Q = 2C$

W

$$V = \frac{W}{Q}$$

$$W = V \times Q$$

$$= 12 \times 2 = 24 J$$

[NCERT]

Q. How much energy is given to each coulomb of charge passing through a 6 V battery?

6 J

Q. What is the potential difference if 30 J of work is done to move 5 C of charge?

a) 5 V

b) 6 V

c) 150 V

d) 25 V

$$V = \frac{W}{Q} = \frac{30}{5} = 6V$$

Q. (a) What is meant by saying that the potential difference is 10 V?

(b) How much work is done in moving a charge of 4 C across two points having 10 V potential difference?

(c) Name the device used to measure this quantity and state how it is connected in a circuit.

Voltmetre

$V = \frac{W}{Q} = 10J$

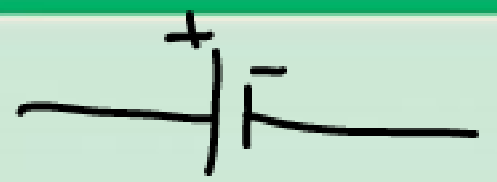
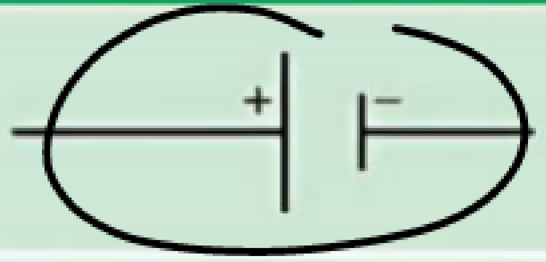
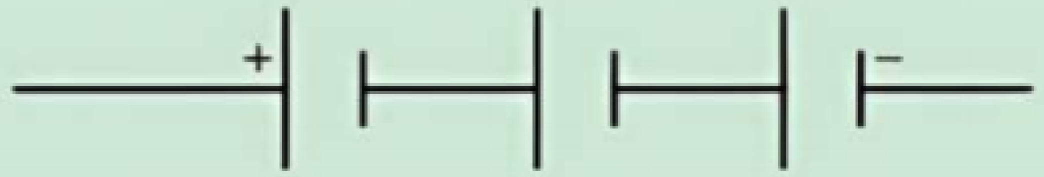
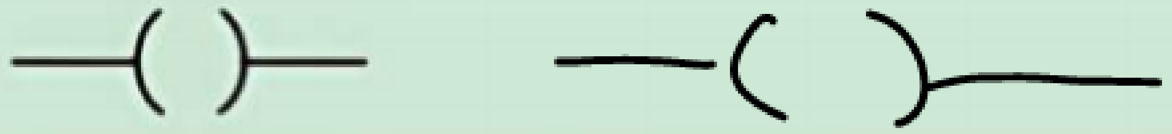
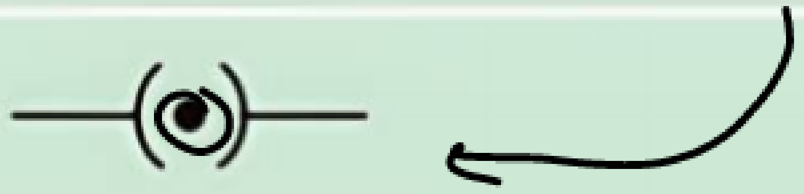
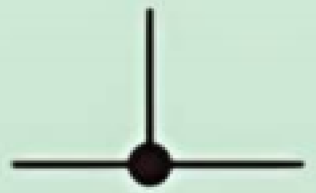
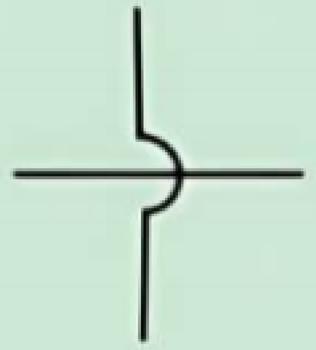
10V

ELECTRICAL CIRCUIT

Electrical circuit is a closed path of electrical components through which the electric current flows.

An electric circuit consists of electric devices, a source of energy and wires that are connected with the help of a switch.

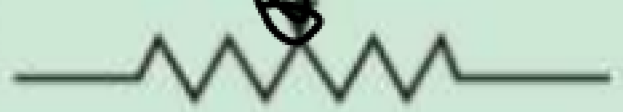
SYMBOLS OF COMPONENTS USED IN CIRCUIT DIAGRAMS

Sl. No.	Components	Symbols
1	An electric cell 	
2	A battery or a combination of cells	
3	Plug key or switch (open)	
4	Plug key or switch (closed)	
5	A wire joint	
6	Wires crossing without joining	

Electric Cell

Battery

Plug Key

Sl. No.	Components	Symbols
7	Electric bulb	 or 
8	A resistor of resistance R	
9	Variable resistance or rheostat	 or 
10	Ammeter	
11	Voltmeter	

Resistor

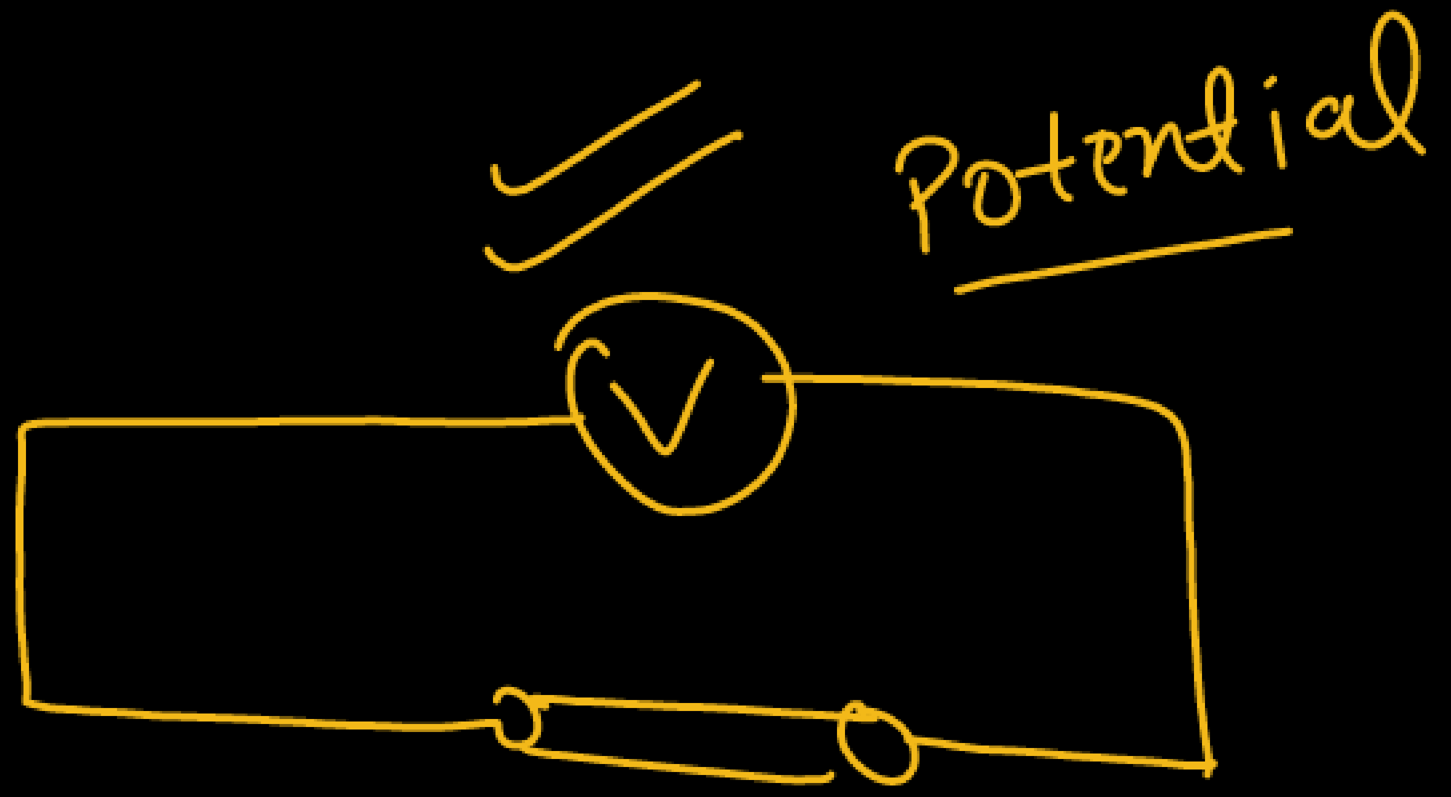
Rheostat

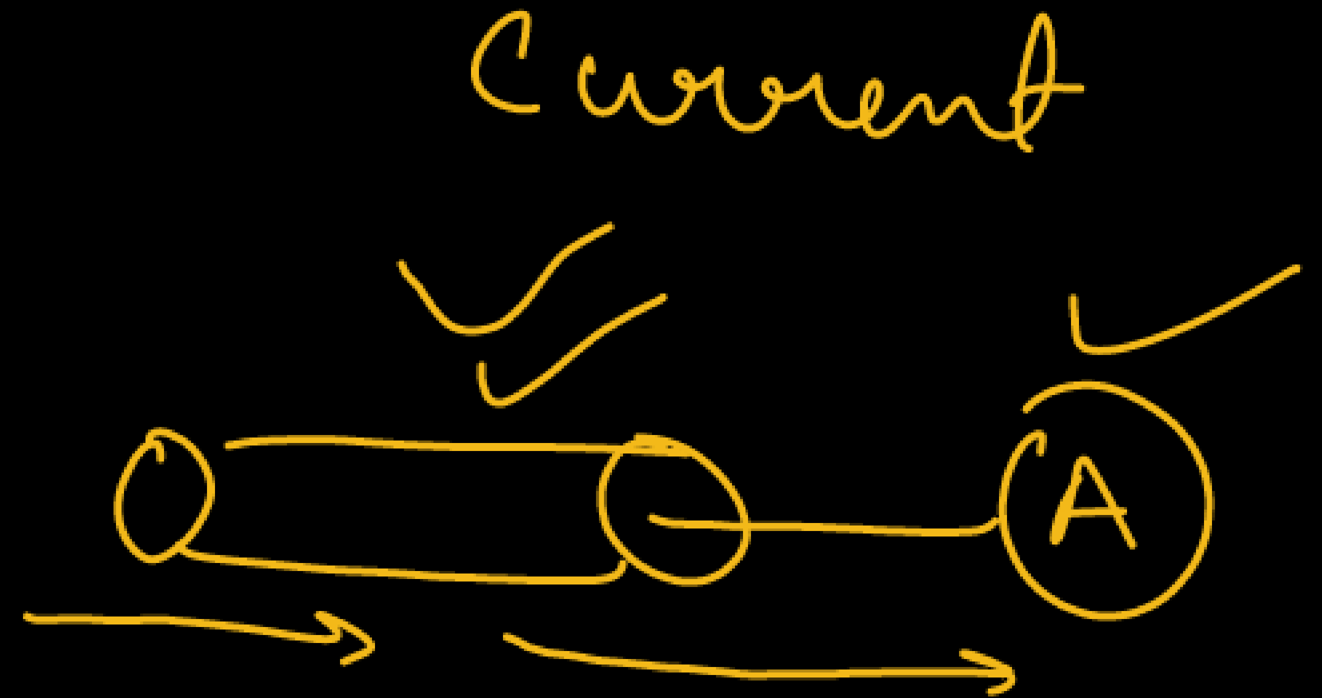
Ammeter

Voltmeter

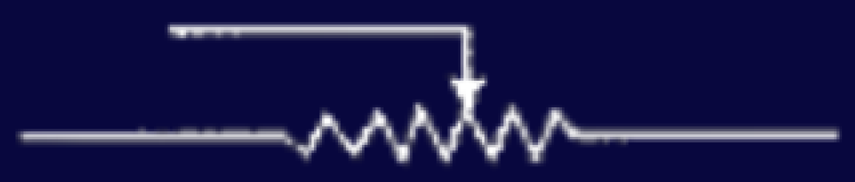
AMMETER AND VOLTEMTER

Ammeter	Voltmeter
Instrument used to measure **electric current** in a circuit.	Instrument used to measure **potential difference (voltage)** between two points in a circuit.
Unit of current is **ampere (A)**.	Unit of potential difference is **volt (V)**.
Always connected **in series** with the circuit component.	Always connected **in parallel** across the circuit component.
Reason: In series, the **same current flows** through the ammeter and the component.	Reason: In parallel, it measures the **exact potential difference** across that component.
Symbol: A circle with **A** inside.	Symbol: A circle with **V** inside.


Potential
V

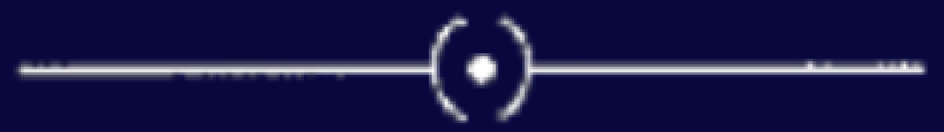

Current
A

Q. Identify the following symbols of commonly used components in a circuit diagram:



(a)

closed switch



(b)

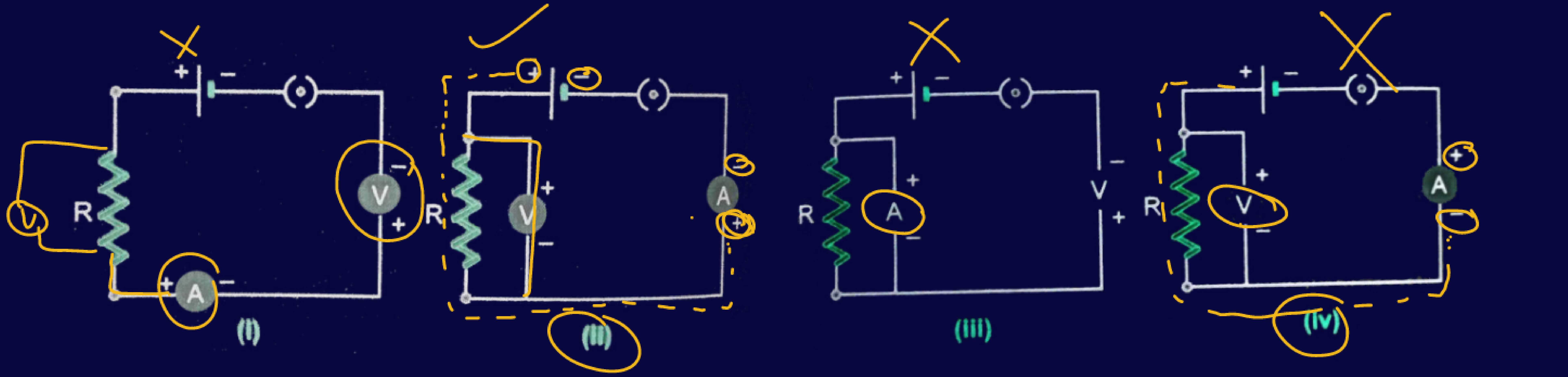
Q. Identify the circuit figure in which the electrical components have been properly connected.

(a) (i)

(b) (ii)

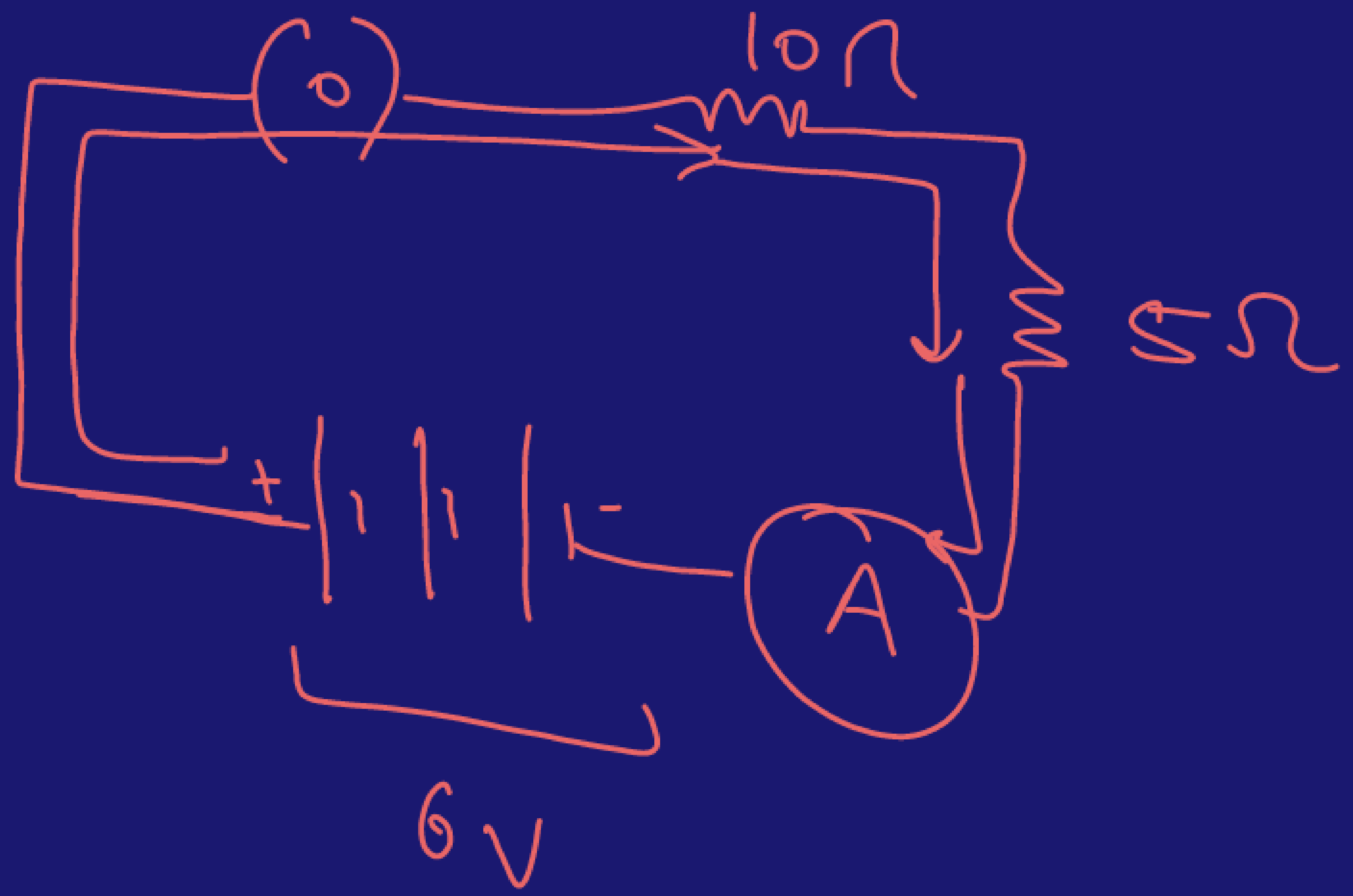
(c) (iii)

(d) (iv)



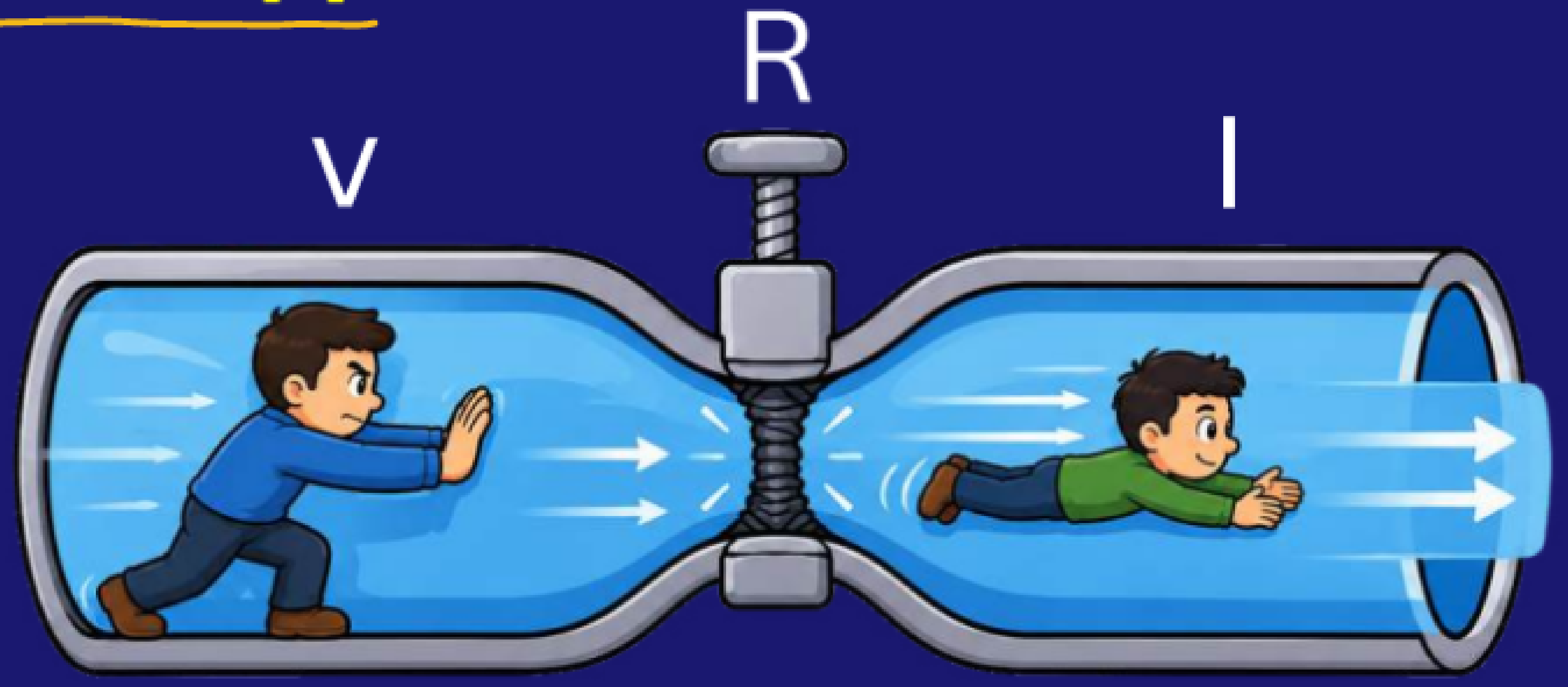
Q. Draw a schematic diagram of a circuit consisting of a battery of 3 cells of 2 V each, a 5 Ω resistor, a 10 Ω resistor, an ammeter, and a plug key, all connected in series.

HW



RESISTANCE

Resistance is the property of a conductor by which it opposes or resists the flow of electric charges through it.



$R = V/I$

SI Unit: Ohm (Ω). Ω

Measuring device - Ohmmeter

Ω → Ohm

1 Ohm: Resistance of a conductor is 1 Ω if a potential difference of 1 V across its ends allows a current of 1 A to flow.

$$1\,\Omega = 1\,V / 1\,A$$

A rheostat is a variable resistor used to regulate the current in an electric circuit without changing the voltage.

OHM'S LAW

$V \propto I$

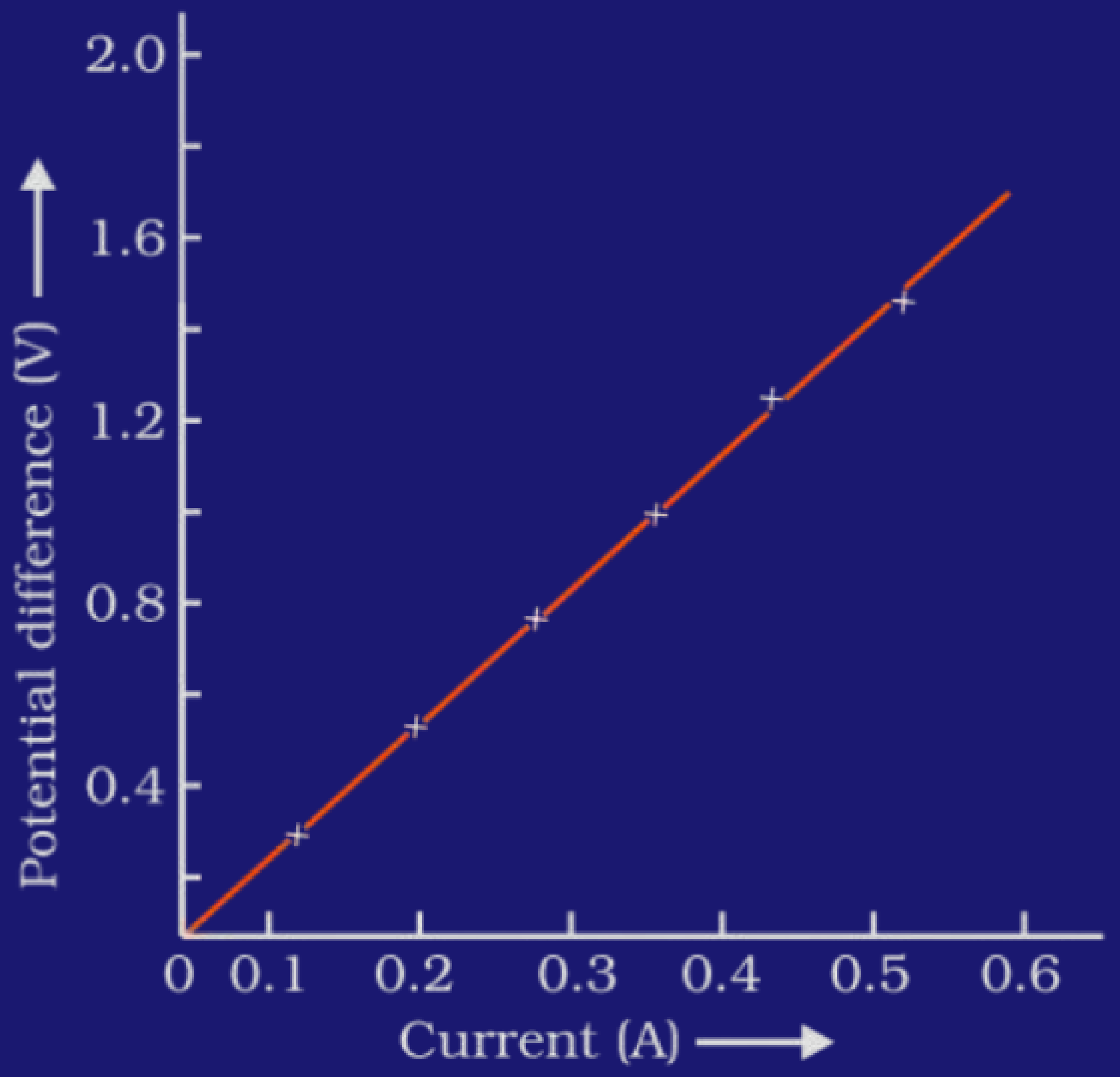
$\boxed{V = IR}$

The potential difference across the ends of a given metallic conductor is directly proportional to the current flowing through it, provided its temperature remains constant.

$$V \propto I$$

$$\boxed{V = IR}$$

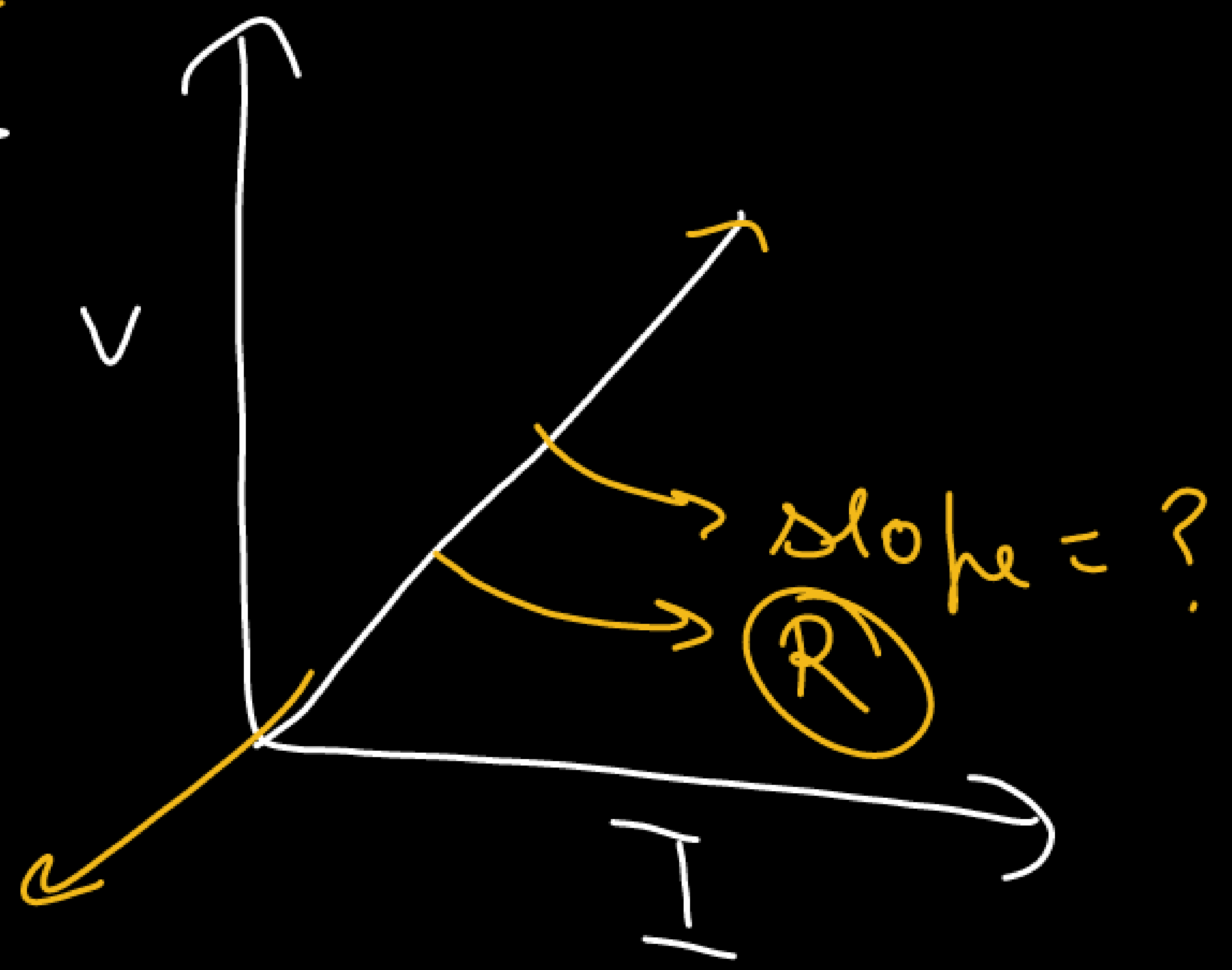
Resistance



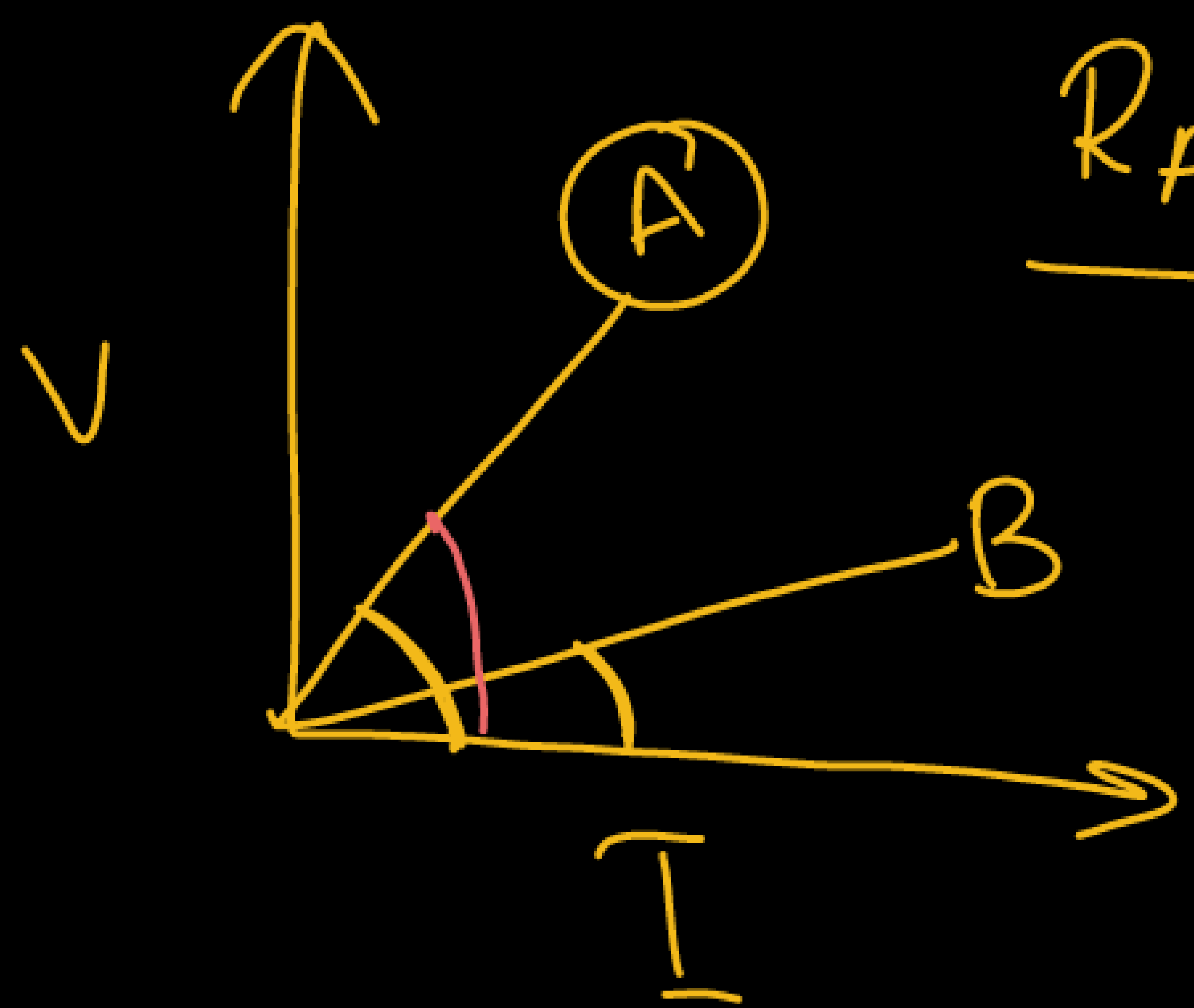
$$V = IR$$

$$\frac{V}{I} = R$$

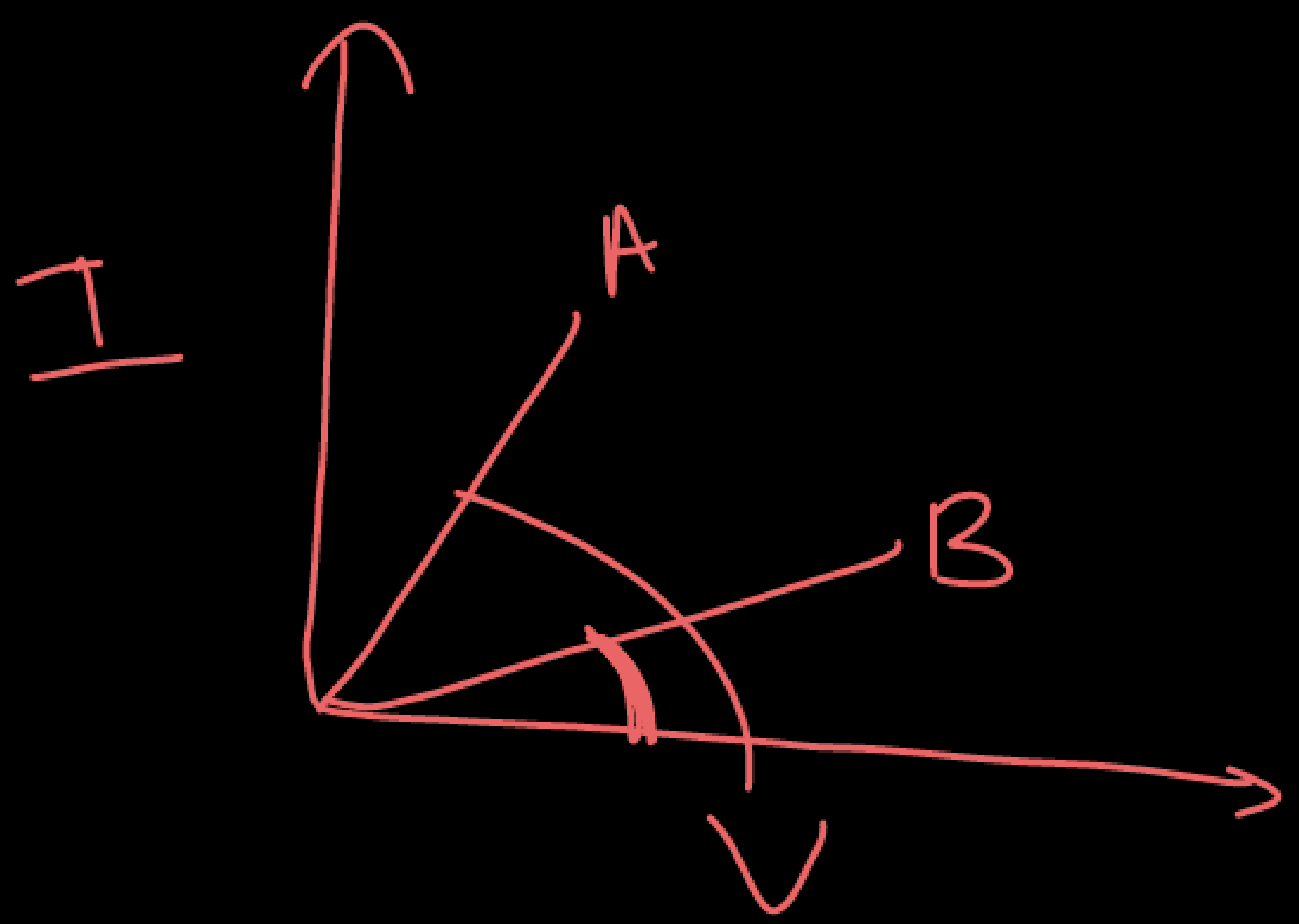
$$\uparrow V \propto I \uparrow$$



$$\frac{V}{I} = R$$



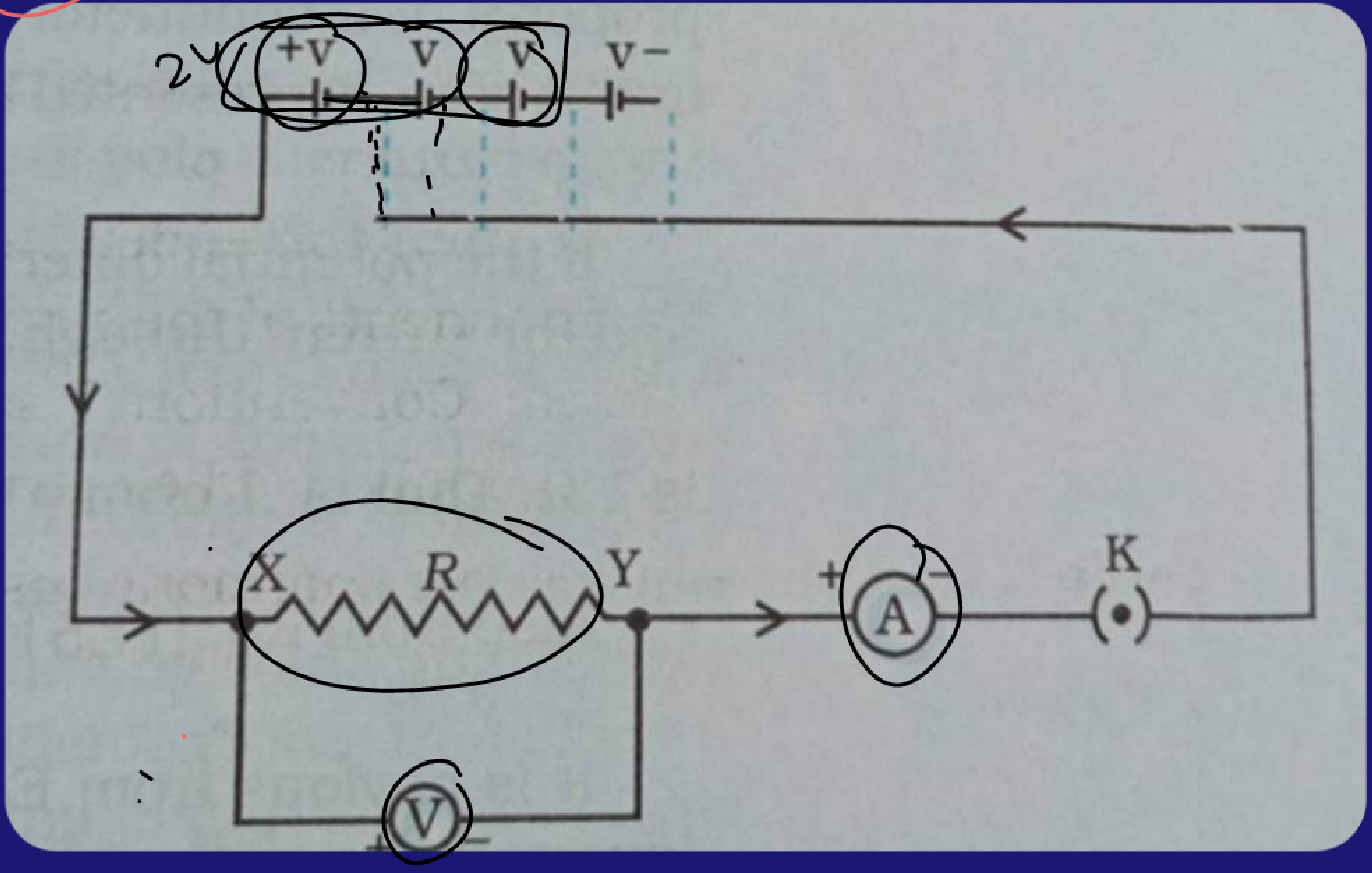
$\underline{R_A > R_B}$



$\underline{R_B > R_A}$

2 Year ✓

ACTIVITY 11.1 – OHM'S LAW



V	I
2V	1A
4V	2A
6V	3A

$\downarrow T V \alpha I T \downarrow$

Q. A student carries out an experiment and plots the V-I graph of three samples of nichrome wire with resistances R_1, R_2 and R_3 respectively. Which of the following is true?

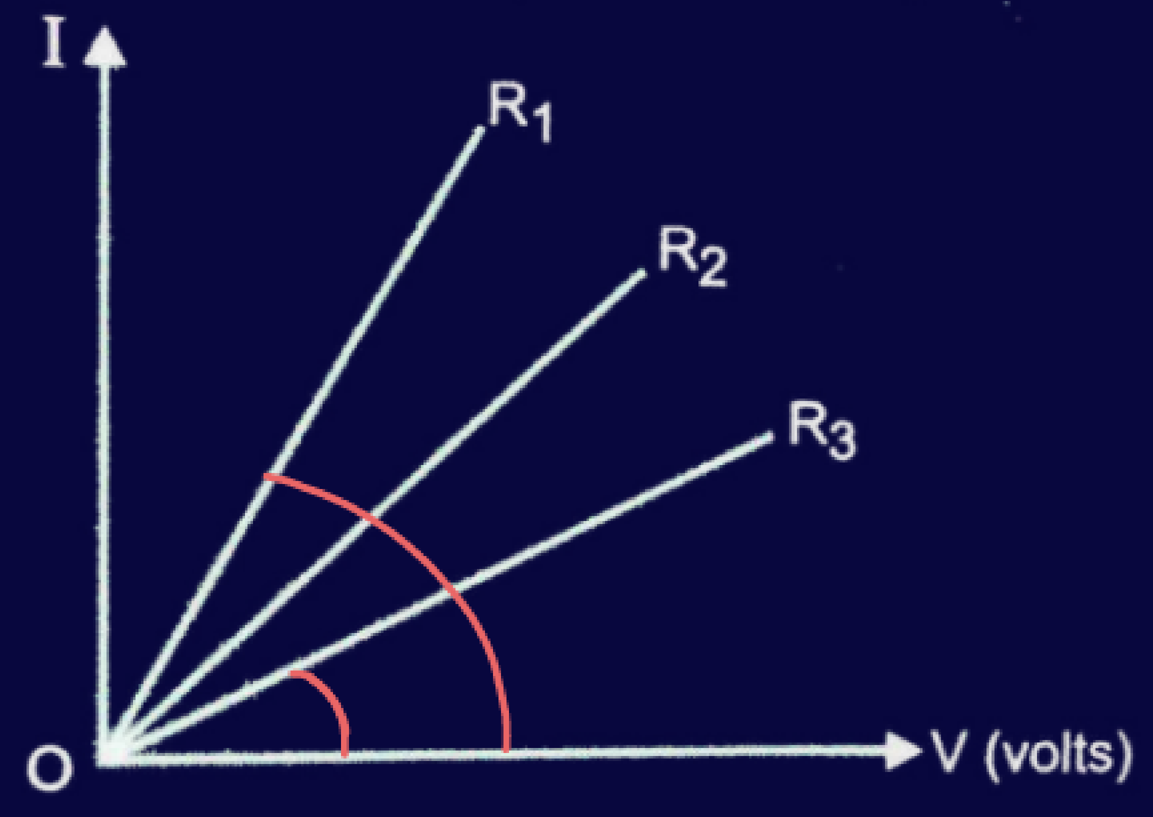
(a) $R_1 = R_2 = R_3$

(b) $R_1 > R_2 > R_3$

(c) $R_3 > R_2 > R_1$

(d) $R_2 > R_3 > R_1$

$R_3 > R_2 > R_1$



[NCERT]

Q. Let the resistance of an electrical component remains constant while the potential difference across the two ends of the component decreases to half of its former value. What change will occur in the current through it?

$V \rightarrow \frac{V}{2}$ ✓

$V \propto I$

$I \rightarrow \frac{I}{2}$

According to Ohm's law,

I = V/R

Here, resistance R remains constant.

So, current I is directly proportional to potential difference V.

If potential difference decreases to half, then current will also decrease to half of its former value.

Therefore, the current through the component becomes half.

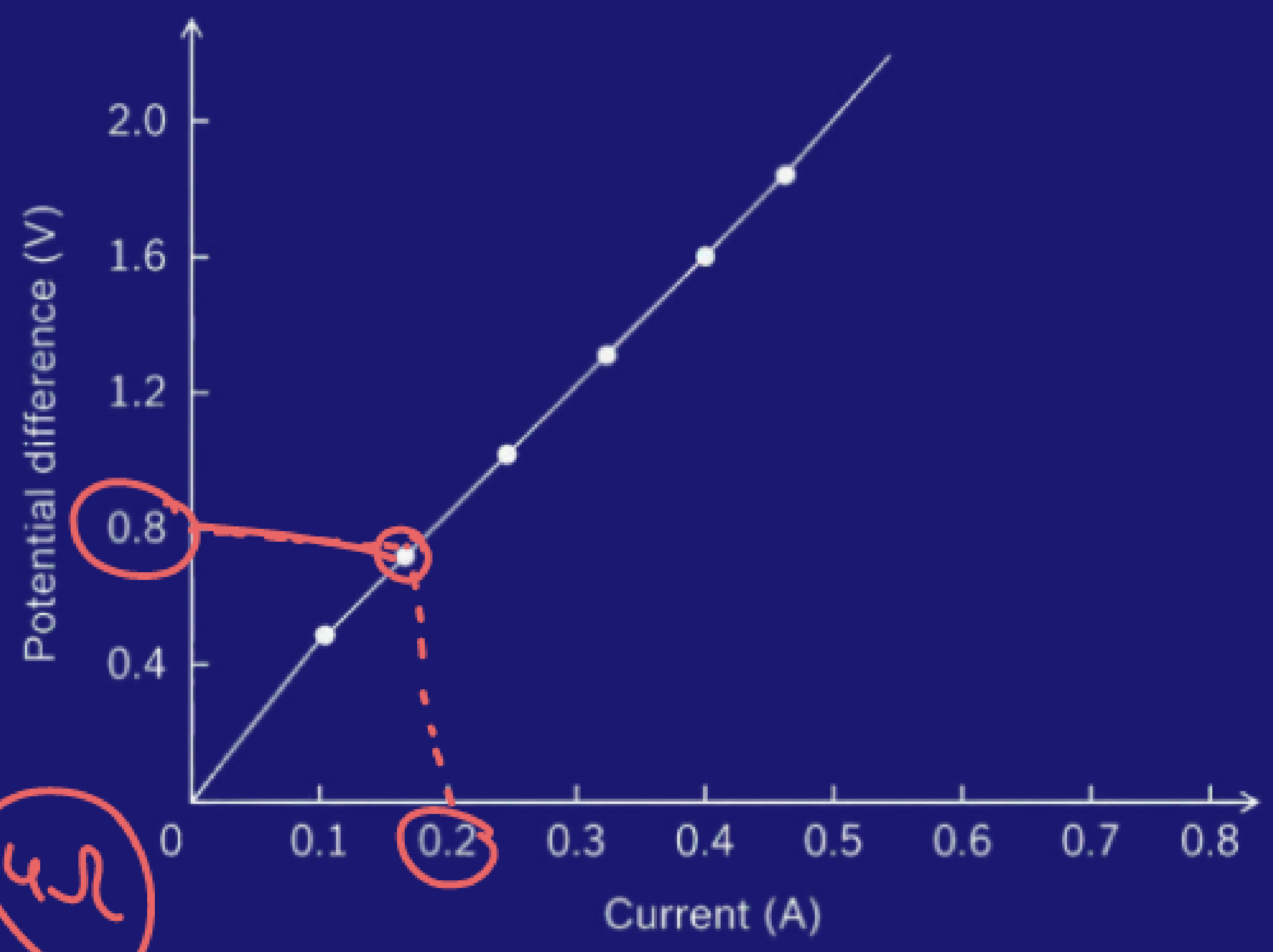
[NCERT]

Q. A V-I graph for a nichrome wire is given below. What do you infer from this graph? Draw a labelled circuit diagram to obtain such a graph.

$V \propto I$

$R = \frac{V}{I}$

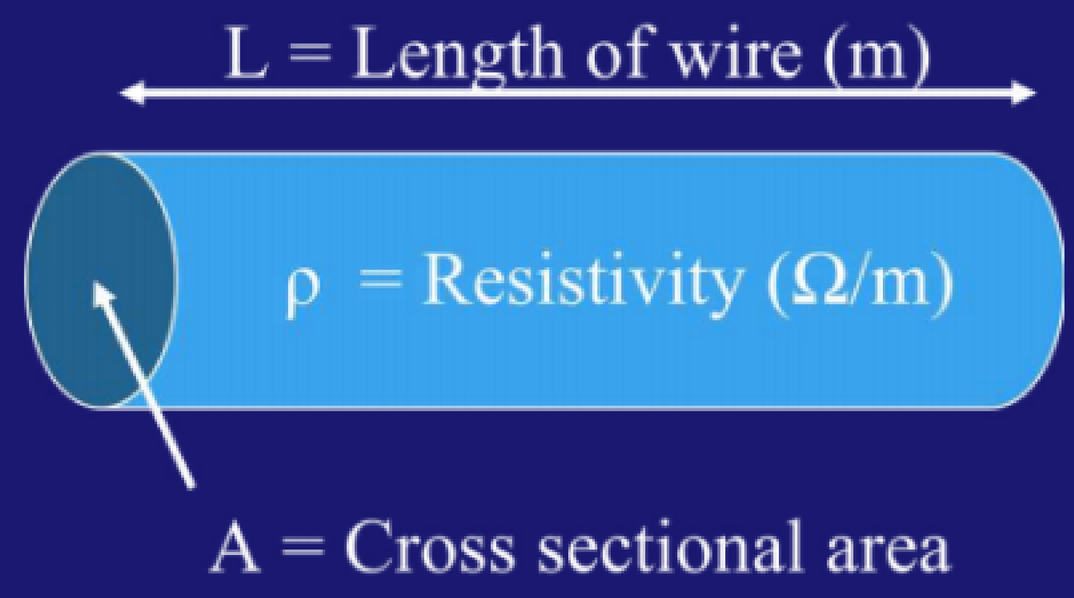
$R = \frac{V}{I} = \frac{0.8}{0.2} = 4\Omega$



FACTORS AFFECTING RESISTANCE

(i) Length of the conductor (l): $R \propto L$

Longer the wire → more resistance.



(ii) Inversely proportional to the area of cross-section: $R \propto 1/A$

Thicker wire (larger area) → less resistance.

$R \propto T$

(iii) Directly proportional to the temperature: $R \propto$ Temperature

- **Metals like copper and aluminium have low resistance → good conductors.**
- **Materials like rubber and plastic have very high resistance → insulators.**
- **Some materials like semiconductors (silicon, germanium) have resistance values between conductors and insulators.**

ρ ρ $\boxed{R \propto \rho}$

(iv) Depends on nature of material: (ρ)

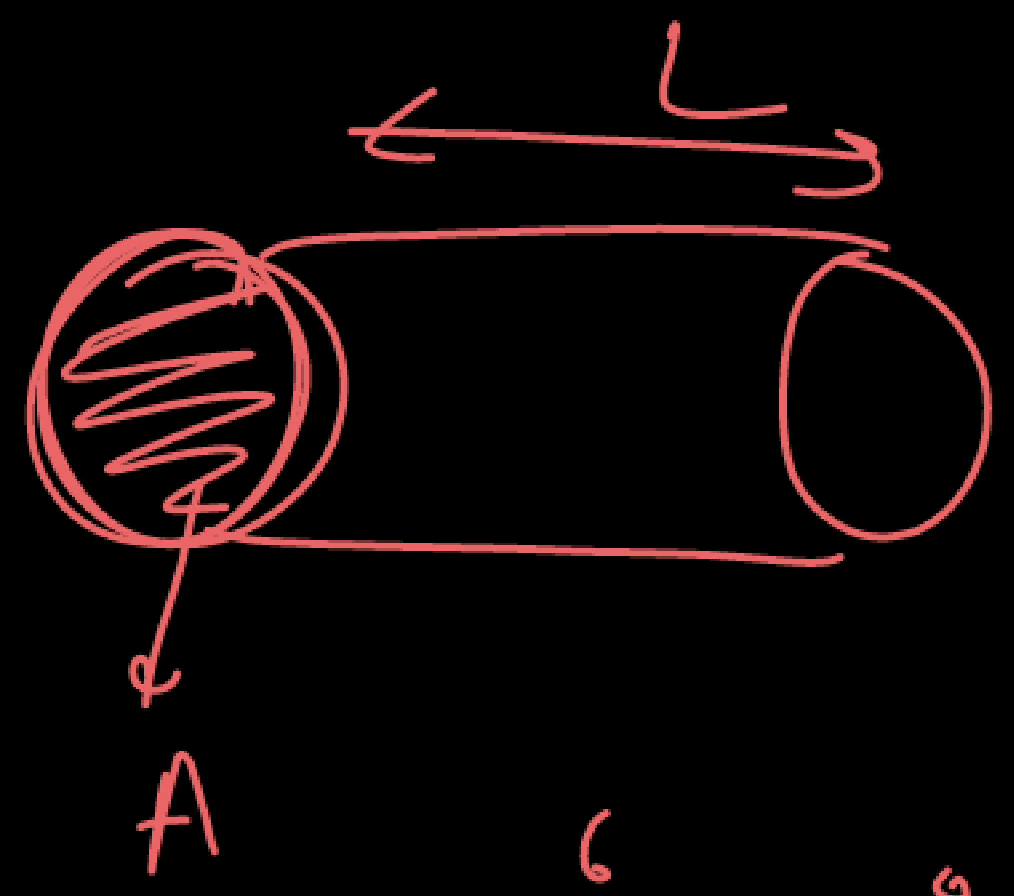
- **Different materials have different $\boxed{\text{resistivity}}$ (ρ).**
- **Example: Copper has very low resistivity, nichrome has higher resistivity.**

$R \propto L$

$R \propto \frac{1}{A}$

$R \propto T$

$R \propto \rho$



'Rho'

$$\boxed{R = \frac{\rho L}{A}}$$

HW

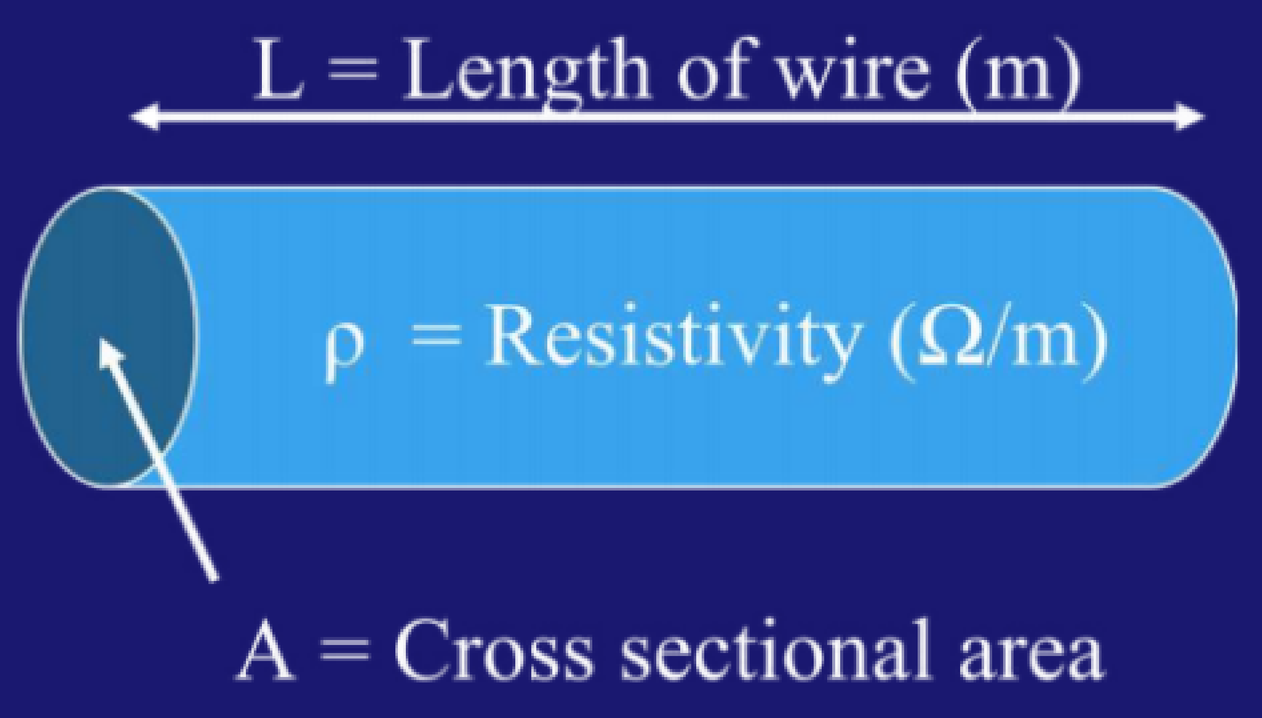
Derive Unit of Resitivity?

$$\boxed{R = \frac{\rho L}{A}}$$

RESISTIVITY

- **Resistance of a conductor is directly proportional to its length, inversely proportional to its area of cross-section, and depends on the material of the conductor. It is denoted by ρ.**

$$R = \rho L/A$$
$$\rho = RA/L$$
$$\rho = \Omega m$$



- ***Resistivity does not change with change in length or area of cross-section but it changes with change in temperature.***

  - Tungsten → filament of bulbs (high melting point).
  - Copper, aluminium → transmission wires (low resistivity).